NOTES FOR GEOMETRIC REPRESENTATION THEORY 1

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The goal of this course is to study certain representations of complex semisimple Lie algebras, such as $\mathfrak{sl}_2(\mathbb{C})$, via geometric methods.

1. Introduction

1.1. **Recollection: semisimple Lie algebras and root systems.** Let $\mathfrak{g}$ be a complex semisimple Lie algebra. We exclude the case $\mathfrak{g} = 0$ to avoid some awkward words.

We fix a Borel subalgebra $\mathfrak{b} \subseteq \mathfrak{g}$ and a Cartan subalgebra $\mathfrak{h}$ contained in $\mathfrak{b}$. Let $\mathfrak{b}^-$ be the opposite Borel subalgebra such that $\mathfrak{b}^- \cap \mathfrak{b} = \mathfrak{h}$. Write $\mathfrak{n} := [\mathfrak{b}, \mathfrak{b}]$ and $\mathfrak{n}^- := [\mathfrak{b}^-, \mathfrak{b}^-]$.

Recall the following standard terminologies and facts.

- Elements in $\mathfrak{h}^* := \mathsf{Hom}_{\mathbb{C}}(\mathfrak{h}, \mathbb{C})$ are called *weights*.
- Nonzero weights that appear in $\mathfrak{g}$ are called *roots*. We have a *root decomposition*
$$\mathfrak{g} \simeq \mathfrak{h} \bigoplus (\bigoplus_{\alpha \in \Phi} \mathfrak{g}_\alpha),$$
where Φ is the set of roots. Each root subspace $\mathfrak{g}_\alpha$ is 1-dimensional.
- Roots that appear in $\mathfrak{b}$ are called *positive roots*. We have $\Phi = \Phi^+ \sqcup -\Phi^+$.
- A positive root is *simple* if it is not the sum of other positive roots. Let $\Delta \subseteq \Phi$ be the subset of positive roots.
- Let $\mathfrak{h}^*_{\mathbb{R}} := \mathbb{R}\Phi \subseteq \mathfrak{h}^*$ be the $\mathbb{R}$-span of Φ. This gives a real structure on $\mathfrak{h}^*$. In other words, we have $\mathfrak{h}^*_{\mathbb{R}} \otimes_{\mathbb{R}} \mathbb{C} \xrightarrow{\simeq} \mathfrak{h}^*$.
- The *Killing form* of $\mathfrak{g}$ restricts to a non-degenerate symmetric bilinear form on $\mathfrak{h}$. Let $(-,-) : \mathfrak{h}^* \times \mathfrak{h}^* \to \mathbb{C}$ be the dual form. Its restriction on $\mathfrak{h}^*_{\mathbb{R}}$ is an inner form.
- The pair $(\mathfrak{h}^*_{\mathbb{R}}, \Phi)$ is a *root system*. Let W be the *Weyl group* of this root system, which is the reflection group acting on $\mathfrak{h}^*_{\mathbb{R}}$ with generators
$$s_\alpha(\lambda) := \lambda - 2\frac{(\alpha, \lambda)}{(\alpha, \alpha)}\alpha, \quad \alpha \in \Phi$$
- Elements $s_\alpha \in W$ ($\alpha \in \Delta$) are called *simple reflections*. Recall that W is generated by simple reflections. For $w \in W$, the *length* $l(w)$ of w is the minimal length of writing w as a product of simple reflections. There is a unique longest element in W, which we denote by w_0.
- For each root $\alpha \in \Phi$, we have the corresponding *coroot* $\check{\alpha} \in \mathfrak{h}$ characterized by the formula
$$\check{\alpha}(\lambda) = 2\frac{(\alpha, \lambda)}{(\alpha, \alpha)}, \quad \forall \lambda \in \mathfrak{h}^*.$$
- A weight $\lambda \in \mathfrak{h}^*$ is *integral* if $\check{\alpha}(\lambda) \in \mathbb{Z}$ for any root $\alpha \in \Phi$. Integral weights form a $\mathbb{Z}$-lattice $\Lambda \subseteq \mathfrak{h}^*$. In other words, we have $\Lambda \otimes_{\mathbb{Z}} \mathbb{C} \xrightarrow{\simeq} \mathfrak{h}^*$.
- An *integral* weight $\lambda \in \mathfrak{h}^*$ is *dominant* if $\check{\alpha}(\lambda) \geq 0$ for any positive root $\alpha \in \Phi^+$. Dominant integral weights form a semigroup $\Lambda^+ \subseteq \Lambda$.
- For weights $\lambda, \lambda' \in \Lambda$, we say $\lambda \leq \lambda'$ if $\lambda' - \lambda$ is *positive*. In other words, $\lambda' - \lambda$ is a finite sum of positive roots.
- Let
$$\rho := (\sum_{\alpha \in \Phi^+} \alpha)/2$$

be the half sum of all positive roots. Recall that $\check{\alpha}(\rho) = 1$ for any simple positive root $\alpha \in \Delta$.

1.2. **From finite dimentional representations to Verma modules.** Recall the following well-known result.

Theorem 1.2.1. *The category $\mathfrak{g}\text{–}\mathsf{mod}_{\mathsf{fd}}$ of finite dimensional $\mathfrak{g}$-modules is semisimple. Moreover, there is a canonical bijection*

$$\Lambda^+ \xrightarrow{\simeq} \mathsf{Irr}(\mathfrak{g}\text{–}\mathsf{mod}_{\mathsf{fd}}),\ \lambda \mapsto L_\lambda \tag{1.1}$$

such that L_λ has highest weight λ.

As an $\mathfrak{h}$-modules, we have

$$\bigoplus_{\mu \leq \lambda, \mu \in \Lambda} \mathbb{C}_\mu \otimes \mathsf{Hom}_{\mathfrak{h}}(\mathbb{C}_\mu, L_\lambda) \xrightarrow{\simeq} L_\lambda,$$

where each direct summand in the LHS is a *weight subspace* of L_λ. The following question is natural.

Question 1.2.2. How to calculate the dimension of the μ-weight subspace of L_λ, i.e., $\dim(\mathsf{Hom}_{\mathfrak{h}}(\mathbb{C}_\mu, L_\lambda))$?

Example 1.2.3. For $\mathfrak{g} = \mathfrak{sl}_2$, let $\mathfrak{b}$ and $\mathfrak{h}$ be its standard Borel and Cartan subalgebra. We have short exact sequences

$$0 \to \mathfrak{h} \to \mathbb{C}^2 \xrightarrow{\Sigma} \mathbb{C} \to 0$$

and

$$0 \to \mathbb{C} \xrightarrow{\Delta} \mathbb{C}^2 \to \mathfrak{h}^* \to 0.$$

We identify $\mathfrak{h}^*$ with $\mathbb{C}$ such that $(\lambda_1, \lambda_2) \in \mathbb{C}^2$ is sent to $l := \lambda_2 - \lambda_1$ by the map $\mathbb{C}^2 \to \mathfrak{h}^* \simeq \mathbb{C}$. Note that:

- A weight $\lambda \in \mathfrak{h}^*$ is integral iff the corresponding number l is integral;
- A weight $\lambda \in \mathfrak{h}^*$ is dominant iff the corresponding number l is non-negative;
- For weights $\lambda, \lambda' \in \mathfrak{h}^*$ and the corresponding numbers l, l', we have $\lambda \leq \lambda'$ iff $l' - l \in 2\mathbb{Z}$.

The vector space L_λ can be identified with the space of homogeneous polynomials $f \in \mathbb{C}[x_1, x_2]$ with degree l, and the $\mathfrak{sl}_2$-action on this vector space is given by changing coordinates. It follows that weights of L_λ correspond to numbers $l - 2d$, $d = 0, \cdots, l$, and each weight subspace is 1-dimensional. Namely, $x_1^{l-d} x_2^d$ is a generator to the weight subspace corresponding to $l - 2d$.

To describe the solution for general $\mathfrak{g}$, we introduce the *character* of $V \in \mathfrak{g}\text{–}\mathsf{mod}_{\mathsf{fd}}$ by the formula

$$\mathsf{ch}(V) := \sum_{\mu \in \mathfrak{h}^*} \dim(\mathsf{Hom}_{\mathfrak{h}}(\mathsf{C}_\mu, V)) T^\mu,$$

which is an element in the polynomial algebra. Note that we have

$$\mathsf{ch}(V \bigoplus V') = \mathsf{ch}(V) + \mathsf{ch}(V'),\ \ \mathsf{ch}(V \otimes V') = \mathsf{ch}(V) \cdot \mathsf{ch}(V').$$

Theorem 1.2.4 (Weyl character formula)**.** *We have*

$$\mathsf{ch}(L_\lambda) \cdot \sum_{w \in W} (-1)^{l(w)} T^{w(\rho)} = \sum_{w \in W} (-1)^{l(w)} T^{w(\lambda + \rho)}.$$

In other words, we can calculate $\mathsf{ch}(L_\lambda)$ by expanding

$$\sum_{w\in W}(-1)^{l(w)}\frac{T^{w(\lambda+\rho)}}{\sum_{w'\in W}(-1)^{l(w')}T^{w'(\rho)}}, \tag{1.2}$$

as a *formal infinite* sum of T^μ ($\mu\in\Lambda$), which turns out to be a finite sum.

Remark 1.2.5. Note that the denominator does not depend on λ.

The following example shows that each term in the alternating sum (1.2) is a genuine infinite sum.

Example 1.2.6. For $\mathfrak{g}=\mathfrak{sl}_2$, we have

$$\mathsf{ch}(L_\lambda)=\frac{T^{l+1}-T^{-l-1}}{T-T^{-1}}=T^l+T^{l-2}+\cdots T^{-l}.$$

However, both

$$\frac{T^{l+1}}{T-T^{-1}}=T^l+T^{l-2}+\cdots$$

and

$$\frac{T^{-l-1}}{T-T^{-1}}=T^{-l-2}+T^{-l-4}+\cdots$$

are infinite sums.

This raises the following natural question.

Question 1.2.7. Is there a representation theoretic intepretation for the term

$$\frac{T^{w(\lambda+\rho)}}{\sum_{w\in W}(-1)^{l(w)}T^{w(\rho)}}?$$

For example, for $\mathfrak{g}=\mathfrak{sl}_2$, we can ask the following question.

Question 1.2.8. Can we find (infinite dimensional) $\mathfrak{sl}_2$-modules $N\subseteq M$ with

$$\mathsf{ch}(M)=\frac{T^{l+1}}{T-T^{-1}},\ \ \mathsf{ch}(N)=\frac{T^{-l-1}}{T-T^{-1}}$$

such that $L_\lambda\simeq M/N$?

It turns out there is a positive answer to Question 1.2.7.

Theorem 1.2.9 (Bernstein–Gelfand–Gelfand resolution)**.** *There is a canonical resolution*

$$0\to M_{w_0\cdot\lambda}\cdots\to\bigoplus_{\ell(w)=2}M_{w\cdot\lambda}\to\bigoplus_{\ell(w)=1}M_{w\cdot\lambda}\to M_\lambda\to L_\lambda\to 0.$$

In above, $M_{w\cdot\lambda}$ is the *Verma module* with highest weight

$$w\cdot\lambda:=w(\lambda+\rho)-\rho,$$

which is an *infinite-dimensional* $\mathfrak{g}$-module.

The upshot is: we need to study infinite-dimensional representations to understand the finite-dimensional ones better.

Warning 1.2.10 (or features)**.** The category $\mathfrak{g}\text{–}\mathsf{mod}$ of (all) $\mathfrak{g}$-modules is not semisimple. A $\mathfrak{g}$-module may fail to be integrable.

Example 1.2.11. As we will soon see, for $\mathfrak{g}=\mathfrak{sl}_2$, the short exact sequence

$$0\to M_{w_0\cdot\lambda}\to M_\lambda\to L_\lambda\to 0$$

does not split. Also, M_λ is not an (algebraic) representation for SL_2.

1.3. **Recollection: enveloping algebra.** Recall that for any Lie algebra $\mathfrak{g}$, we have a canonical equivalence

$$\mathfrak{g}\text{–}\mathsf{mod} \simeq U(\mathfrak{g})\text{–}\mathsf{mod}, \tag{1.3}$$

where $U(\mathfrak{g})$ is the *universal enveloping algebra* of $\mathfrak{g}$, which is an associative (but not commutative) $\mathbb{C}$-algebra. Namely, let

$$T(\mathfrak{g}) := \bigoplus_{i=0}^{\infty} \mathfrak{g}^{\otimes i}$$

be the tensor algebra for the underlying vector space $\mathfrak{g}$. Consider the two-sided ideal $I \subseteq T(\mathfrak{g})$ generated by elements

$$X \cdot Y - Y \cdot X - [X, Y] \in T(\mathfrak{g}).$$

One can check that

$$U(\mathfrak{g}) := T(\mathfrak{g})/I$$

fits into the desired equivalence (1.3).

For future reference, let us recall that the (non-negative) grading on $T(\mathfrak{g})$ induces the *PBW filtration* on $U(\mathfrak{g})$:

$$\mathsf{F}^{\leq n} U(\mathfrak{g}) := \mathrm{Im}(\bigoplus_{i=0}^{n} \mathfrak{g}^{\otimes i} \to U(\mathfrak{g})).$$

One can check that this makes $U(\mathfrak{g})$ an *almost commutative filtered associative algebra*, i.e.,

$$\mathsf{F}^{\leq m} U(\mathfrak{g}) \cdot \mathsf{F}^{\leq n} U(\mathfrak{g}) \subseteq \mathsf{F}^{\leq m+n} U(\mathfrak{g})$$

and

$$[\mathsf{F}^{\leq m} U(\mathfrak{g}), \mathsf{F}^{\leq n} U(\mathfrak{g})] \subseteq \mathsf{F}^{\leq m+n-1} U(\mathfrak{g}).$$

In particular,

$$\mathsf{gr}^{\bullet} U(\mathfrak{g}) := \bigoplus_{n=0}^{\infty} \mathsf{gr}^{n} U(\mathfrak{g}) := \bigoplus_{n=0}^{\infty} \mathsf{F}^{\leq n} U(\mathfrak{g}) / \mathsf{F}^{\leq n-1} U(\mathfrak{g})$$

is a graded commutative algebra.

Theorem 1.3.1 (Poincare–Birkhoff–Witt)**.** *The map*

$$\mathfrak{g} \to \mathsf{F}^{\leq 1} U(\mathfrak{g}) \to \mathsf{gr}^{1} U(\mathfrak{g}) \to \mathsf{gr}^{\bullet} U(\mathfrak{g})$$

induces an isomorphism

$$\mathsf{Sym}^{\bullet}(\mathfrak{g}) \xrightarrow{\simeq} \mathsf{gr}^{\bullet} U(\mathfrak{g}).$$

The construction $\mathfrak{g} \mapsto U(\mathfrak{g})$ is functorial. In other words, for any Lie algebra homomorphism $\mathfrak{g} \to \mathfrak{g}'$, we have a ring homomorphism $U(\mathfrak{g}) \to U(\mathfrak{g}')$ such that the following diagram commute

$$\begin{tikzcd}
\mathfrak{g}'\text{–}\mathsf{mod} \arrow[r, "\mathsf{res}"] \arrow[d, "\simeq"] & \mathfrak{g}\text{–}\mathsf{mod} \arrow[d, "\simeq"] \\
U(\mathfrak{g}')\text{–}\mathsf{mod} \arrow[r, "\mathsf{res}"] & U(\mathfrak{g})\text{–}\mathsf{mod}.
\end{tikzcd}$$

It follows that the top horizontal functor admits a left adjoint

$$\mathsf{ind}_{\mathfrak{g}}^{\mathfrak{g}'} : \mathfrak{g}\text{–}\mathsf{mod} \to \mathfrak{g}'\text{–}\mathsf{mod}, \;\; M \mapsto U(\mathfrak{g}') \underset{U(\mathfrak{g})}{\otimes} M$$

Example 1.3.2. When $\mathfrak{g}' = 0$, we obtain a functor

$$\mathfrak{g}\text{–}\mathsf{mod} \to \mathsf{Vect}, \quad M \mapsto \mathbb{C} \underset{U(\mathfrak{g})}{\otimes} M =: M_{\mathfrak{g}}.$$

The vector space $M_{\mathfrak{g}}$ is called the *coinvariance* of M.

1.4. **Verma modules.**

Construction 1.4.1. Consider the following diagram of Lie algebras

$$\begin{tikzcd} & \mathfrak{b} \arrow[dl] \arrow[dr] & & \\ \mathfrak{g} & & \mathfrak{b}/[\mathfrak{b},\mathfrak{b}] & \mathfrak{h}. \arrow[l, "\simeq"'] \end{tikzcd}$$

For any $\lambda \in \mathfrak{h}^*$, by restriction, we view $\mathbb{C}_\lambda$ as a 1-dimensional $\mathfrak{b}$-module. By induction, we obtain a $\mathfrak{g}$-module

$$M_\lambda := \mathsf{ind}_{\mathfrak{b}}^{\mathfrak{g}}(\mathbb{C}_\lambda) = U(\mathfrak{g}) \underset{U(\mathfrak{b})}{\otimes} \mathbb{C}_\lambda.$$

Exercise 1.4.2. (Problem 1, Homework 1) Show that

$$U(\mathfrak{g}) \simeq U(\mathfrak{n}^-) \otimes U(\mathfrak{b})$$

as $(U(\mathfrak{n}^-), U(\mathfrak{b}))$-bimodules. Deduce that

$$M_\lambda \simeq U(\mathfrak{n}^-)$$

as $\mathfrak{n}^-$-module.

Exercise 1.4.3. (Problem 2, Homework 1) Show that M_λ is a semisimple $\mathfrak{h}$-module and each weight subspace is finite-dimensional.

In particular, we can talk about *weights* and *weight subspaces* of M_λ.

Exercise 1.4.4. (Problem 3, Homework 1) Consider the composition

$$\mathbb{C} \xrightarrow{\subseteq} U(\mathfrak{n}^-) \xrightarrow{\simeq} M_\lambda,$$

where the first map is the structure map for the $\mathbb{C}$-algebra $U(\mathfrak{n}^-)$, and the second bijection is that in Exercise 1.4.3. Show that the image of the above composition is the unique highest weight subspace of M_λ.

Definition 1.4.5. We call M_λ the *Verma module* with highest weight λ.

The goal of this course is to study Verma modules and the subcategory $\mathcal{O} \subseteq \mathfrak{g}\text{–}\mathsf{mod}$ generated by them.

Exercise 1.4.6. (Problem 4, Homework 1) Deduce Theorem 1.2.4 from Theorem 1.2.9. You can achieve this *without* using the Weyl denominator formula. Hint: find $\mathsf{ch}(U(\mathfrak{n}^-))$ by studying the case $\lambda = 0$.

1.5. **Geometric incarnations.** Let G be an algebraic group with Lie algebra $\mathfrak{g}$. Let $H \subseteq B \subseteq G$ be the Cartan and Borel subgroups corresponding to $\mathfrak{h} \subseteq \mathfrak{b}$. The quotient G/B is a complex variety, called the *flag variety*.

In future lectures, for each character λ of H (which in particular is an integral weight of $\mathfrak{h}$), we will construct a canonical G-equivariant line bundle $\mathcal{O}(\lambda)$ on G/B. Roughly speaking, we consider the trivial line bundle $\mathcal{O}_G$ on G, equipped with the trivial (G, B)-equivariant structure, but twist the right B-equivariant structure by the character λ. By descent theory, this indeed gives a G-equivariant line bundle on G/B.

Example 1.5.1. For $G = \mathsf{SL}_2$, the standard action of G on $\mathbb{A}^2 \smallsetminus 0$ induces a transitive action of G on $\mathbb{P}^1$. The stablizer of $\infty \in \mathbb{P}^1$ for this action is the standard Borel subgroup $B \subseteq G$. It follows that $G/B \simeq \mathbb{P}^1$.

Via this isomorphism, for a character λ with

$$\lambda(\mathsf{diag}(t, t^{-1})) = t^n,$$

the line bundle $\mathcal{O}(\lambda)$ on G/B corresponds to the line bundle $\mathcal{O}(n)$ on $\mathbb{P}^1$.

Theorem 1.5.2 (Borel–Weil–Bott)**.** *For dominant λ, we have a canonical isomorphism of G-modules*

$$\Gamma(G/B, \mathcal{O}(\lambda)) \simeq L_\lambda. \tag{1.4}$$

This motivates the following question.

Question 1.5.3. Is it possible to realize the Verma module M_λ as the global section of some objects over G/B?

We should not restrict ourselves to vector bundles on G/B because there are very few of them. Experiences in algebraic geometry suggest we should try (quasi-coherent) sheaves. However, we should first answer the following preliminary question.

Question 1.5.4. For a quasi-coherent $\mathcal{O}$-module $\mathcal{F}$ over G/B, is there some additional structure on $\mathcal{F}$ that makes $\Gamma(G/B, \mathcal{F})$ a $\mathfrak{g}$-module?

Note that in (1.4), the G-action on the LHS comes from the G-equivariant structure on $\mathcal{O}(\lambda)$. However, the Verma module M_λ is *never* a G-module.

Exercise 1.5.5. (Problem 5, Homework 1) Show that M_λ is not an algebraic representation of G.

Nevertheless, there is a beautiful answer Qusetion 1.5.4, motivated by the following observation.

Theorem 1.5.6. *Let $T_{G/B}$ be the tangent bundle on G/B. We have a canonical isomorphism of Lie algebras:*

$$\Gamma(G/B, T_{G/B}) \simeq \mathfrak{g}.$$

In other words, the Lie algebra of global vector fields on G/B is isomorphic to $\mathfrak{g}$.

We will prove Theorem 1.5.6 in future lectures. For now, please convince yourself using the SL_2-case.

Exercise 1.5.7. (Problem 6, Homework 1) Find an isomorphism of Lie algebras

$$\Gamma(\mathbb{P}^1, T_{\mathbb{P}^1}) \simeq \mathfrak{sl}_2.$$

Theorem 1.5.6 suggests that we should consider sheaves $\mathcal{F}$ over G/B equipped with an action of the tangent sheaf $T_{G/B}$. One can call them *sheaves with connections*, but the standard terminology is *D-modules*. Here the letter "D" stands for "differential".

In future lectures, for each Verma module M_λ, we will attach a (twisted) D-module $\mathcal{F}_\lambda$ on G/B such that

$$\Gamma(G/B, \mathcal{F}_\lambda) \simeq M_\lambda$$

as a $\mathfrak{g}$-modules. Moreover, the BGG resolution (Theorem 1.2.9) will be proved by finding a resolution of $\mathcal{O}(\lambda)$ by the (twisted) D-modules $\mathcal{F}_{w\cdot\lambda}$, which itself is a sheaf theoretic incarnation of the Bruhat decomposition

$$G/B = \bigcup_{w\in W} BwB/B.$$

As a consequence, we obtain a *geometric* intepretation of the Weyl character formula.

The ultimate goal of this course is to relate $\mathfrak{g}$-modules with D-modules on G/B. This is known as the *localization theory* for semisimple Lie algebras.

2. Category $\mathcal{O}$

2.1. Definition of category $\mathcal{O}$.

Definition 2.1.1. We define BGG's category $\mathcal{O}$ to be the full subcategory of $\mathfrak{g}-\mathsf{mod}$ that contains $\mathfrak{g}$-modules M satisfying the following conditions:

(i) As a $\mathfrak{g}$-module, M is finitely generated.
(ii) As an $\mathfrak{h}$-module, M is semisimple. In other words, M is a *weight module*.
(iii) As an $\mathfrak{n}$-module, M is a union of finite-dimensional submodules.

Lemma 2.1.2. *The Verma module M_λ is contained in $\mathcal{O}$.*

Proof. The Verma module M_λ satisfies condition (i) and (ii) respectively by Exercise 1.4.2 and Exercise 1.4.3.

It remains to prove that M_λ satisfies condition (iii). Let v_λ be a nonzero highest weight vector of M_λ. By Exercise 1.4.4,

$$M_\lambda = \bigcup_i \mathsf{F}^i U(\mathfrak{g}) \cdot v_\lambda.$$

Each $\mathsf{F}^i U(\mathfrak{g}) \cdot v_\lambda$ is finite dimensional. Hence we only need to show these subspaces are $\mathfrak{n}$-stable. For $u \in \mathsf{F}^i U(\mathfrak{g})$ and $x \in \mathfrak{n}$ we have

$$x \cdot (u \cdot v_\lambda) = u \cdot (x \cdot v_\lambda) + [x, u] \cdot v_\lambda.$$

Note that $\mathfrak{h}$ acts on $x \cdot v_\lambda$ with weight strictly greater than λ. Hence $x \cdot v_\lambda = 0$. This implies the claim because

$$[x, u] \in [\mathfrak{g}, \mathsf{F}^i U(\mathfrak{g})] \subset \mathsf{F}^{i-1} U(\mathfrak{g}) \tag{2.1}$$

□

Note that (2.1) also implies the following result.

Lemma 2.1.3. *The $\mathfrak{n}$-action on M_λ is locally nilpotent.*

The following result is obvious.

Lemma 2.1.4. *The subcategory $\mathcal{O} \subseteq \mathfrak{g}-\mathsf{mod}$ is stable under taking finite direct sums and subquotients.*

Corollary 2.1.5. *The category $\mathcal{O}$ is an abelian category.*

Warning 2.1.6. The subcategory $\mathcal{O} \subseteq \mathfrak{g}-\mathsf{mod}$ is *not* stable under taking extensions.

Exercise 2.1.7. (Problem 7, Homework 1) For $\mathfrak{g} = \mathfrak{sl}_2$, find an extension of two Verma modules inside $\mathfrak{g}-\mathsf{mod}$ that is not a weight module. Hint: $\mathsf{ind}_{\mathfrak{b}}^{\mathfrak{g}}$.

Proposition 2.1.8. *Each object in $\mathcal{O}$ is a quotient of a finite successive extension of some Verma modules.*

Proof. Let $M \in \mathcal{O}$ be an object. By condition (i), M is generated as a $\mathfrak{g}$-module by a finite-dimensional subspace $M_0 \subseteq M$. We may enlarge M_0 such that it is a finite dimensional $\mathfrak{h}$-module. In particular, there are finitely many weights in M_0.

Note that each weight in $U(\mathfrak{h}) \cdot M_0 = U(\mathfrak{n}) \cdot M_0$ is greater or equal to a weight in M_0. Since M is a highest weight module, there are finitely many weights satisfying the above property. Since each weight subspace of M is finite-dimensional, we see that $U(\mathfrak{h}) \cdot M_0$ is finite-dimensional. Therefore we can replace M_0 by $U(\mathfrak{h}) \cdot M_0$ and assume M_0 is a $\mathfrak{b}$-module.

Now consider the natural $\mathfrak{g}$-linear map

$$\mathsf{ind}_{\mathfrak{b}}^{\mathfrak{g}}(M_0) \to M \tag{2.2}$$

induced by the $\mathfrak{b}$-linear map $M_0 \to M$. Since M is generated by M_0, the map (2.2) is surjective. Hence we only need to show $\mathsf{ind}_{\mathfrak{b}}^{\mathfrak{g}}(M_0)$ is a successive extension of the Verma modules. For this purpose, we only need to show M_0 is a successive extension of 1-dimensional representations. This follows from the following exercise.

Exercise 2.1.9. Let M_0 be a finite-dimensional $\mathfrak{b}$-module such that the $\mathfrak{h}$-action is semisimple. Show that M_0 is a successive extension of 1-dimensional representations.

□

Corollary 2.1.10. *Let M be an object in $\mathcal{O}$.*

(1) *As an $\mathfrak{n}^-$-module, M is finitely generated.*
(2) *The $\mathfrak{n}$-action on M is locally nilpotent.*
(3) *Each weight subspace of M is finite-dimensional.*

References

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